

Tyler James Haina Ikehara

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Fifth year PhD Candidate whose research interests include Wireless Communications, Electromagnetics, and RF Circuits. Passionate about learning and teaching others about STEM related topics.

Education

Doctor of Philosophy: Electrical Engineering

Graduation Year: 2028

PhD Candidate | **Advisor: Dr. Thomas Marzetta**

University of Maryland College Park – College Park, MD

Master of Science: Electrical Engineering

August 2022 to June 2025

Future Leader Fellow | **Advisor: Dr. Thomas Marzetta**

New York University Tandon School of Engineering – Brooklyn, NY

Relevant Classwork: Linear Systems Approach to Wave Propagation, Probability and Stochastic Processes, Applied Matrix Theory, Machine Learning, Optimization Methods, Digital Signal Processing, Digital Communications, Electromagnetics, Wireless Communications, Linear Algebra I

GPA: 3.945

Bachelor of Science: Electrical Engineering, Minor – Music

September 2018 to June 2022

Summa Cum Laude | *Electrical and Computer Engineering Academic Excellence Award*

Santa Clara University – Santa Clara, CA

Relevant Classwork: Microwave Circuit Analysis, Intro to Communication Systems, Electromagnetics I&II, Linear Algebra, Probability and Statistics, Multivariable Calculus

GPA: 3.976

Research Experience

Conference Talks | Advisor: Dr. Thomas Marzetta

October 2025/October 2026

2026 Asilomar Conference on Signals, Systems and Computers – Pacific Grove, CA

Invited talk as first author titled “Wave Equation Migration for Passive Detection and Localization of Near- and Far-Field Sources”

Focused on the benefits of using the Wave Equation Migration filter for Near-field UWB source localization

2025 Asilomar Conference on Signals, Systems and Computers – Pacific Grove, CA

Presented a talk as first author titled “The Radon Transform, True Time Delay Beamforming, and Ultra-Wideband Antenna Arrays”

Proposed a Radon Transform approach to remove far-field signals received by an Ultra-wideband antenna array in the FR3 band

Graduate Research Assistant | Advisor: Dr. Thomas Marzetta

September 2022 to Present

New York University Tandon School of Engineering – Brooklyn, NY/ University of Maryland College Park – College Park, MD

Researched and proposed a novel electromagnetic feedback receiver array antenna system inspired by the human cochlea

Used linear system theory to use impedance matrix calculations to understand the linear behavior of the proposed system

Research Intern

June 2021 to August 2021

Naval Information Warfare Center Pacific – San Diego, CA

Investigated the effectiveness of Direct Antenna Modulation and Cryocooled circuit components for High Frequency antenna systems

Part of the Naval Research Enterprise Internship Program

Senior Design Project | Advisor: Dr. Kurt Schab

August 2021 to June 2022

Santa Clara University – Santa Clara, CA

Constructed a Radio Telescope using a Software Defined Radio as the main data processing component

Designed a Microstrip Bandpass Filter for the Radio Astronomy frequency band of 1420 MHz

Guided Research | Advisor: Dr. Kurt Schab

July 2020 to August 2021

Santa Clara University – Santa Clara, CA

Researched Software Defined Radios and gained experience using GNU Radio

Implemented a MATLAB script to compare the Signal to Noise Ratio for antennas with varying efficiency and Q-factor

Work Experience

Summer Intern

June 2025 to August 2025

Jet Propulsion Laboratory/California Institute of Technology – Pasadena, CA

Worked on converting simulation models of the Deep Space Network (DSN) downlink receivers from old software to Simulink

Part of the 333E - DSN Communication Ground Systems: Telecommunications Signal Processing group

Graduate Assistant

September 2024 to January 2025

New York University Tandon School of Engineering – Brooklyn, NY

Assisted with grading an undergraduate Linear Systems class

Created quiz and homework solutions and held weekly office hours

Peer Educator

March 2022 to June 2022

Santa Clara University – Santa Clara, CA

Assisted with the Lab section of a Digital Signal Processing class

Answered questions that the students had and ensured that their MATLAB codes were completed correctly

Student Tutor

September 2019 to March 2020

Santa Clara University – Santa Clara, CA

Math Learning Center Peer Tutor – Student Tutoring Program for math at SCU

Ran weekly study groups for students and held one on one tutoring sessions

Financial Modeling Intern

June 2019 to September 2019

Ekahi Health – Honolulu, HI

Worked directly with Robert Walker, the Director of Ekahi Wellness

Created financial models to provide strategies for a Team Based Care collaborative project between Ekahi Wellness and Central Medical Clinic

Peer Tutor

September 2017 to June 2018

Punahou School – Honolulu, HI

Tutored students in various subjects with an emphasis on Math and Science

Assisted teachers during lecture sessions and lab experiments

Skills

Programming: C/MATLAB/Simulink/SCPI/SPICE/GNU Radio/Python/LaTeX/ADS/HFSS

Hardware: Microwave Antenna Equipment, Arduino, Soldering, Software Defined Radio, Analog Computers

Awards/Achievements

SCU Electrical and Computer Engineering Academic Excellence Award

June 2022

SCU Senior Design Best in Session Presentation

June 2022

Tau Beta Pi – California Charter Inductee

November 2020

SCU Dean's List

2018-2021 School Years

Volunteer Work

Trained SCU mentor for the Mentor Collective Program

Assisted Santa Clara University's Math department by interviewing applicant professors.

Volunteer at Sacred Heart Community Service Food Pantry

Punahou Carnival – Food Supply Booth Chair - Managed students and volunteers to keep food booths supplied. The Punahou Carnival is a 2-day scholarship fundraising event that grosses over \$2 million dollars annually.